Analysis of Road Collisions on Public Roads in Great Britain (2024)

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Chapter 1

Introduction

Road accidents are not uncommon incidents in any part of the world. Great Britain is not alien to this fact either. Over the years, the Police have recorded collisions on public roads that were reported. These reports have been recorded from 1979 to recent years (Department for Transport, 2023). From reports compiled over the years, we can observe a downward trend in fatalities. The year 2024 reports the lowest number of fatalities, excluding years affected by the COVID-19 pandemic (Department for Transport, 2025).

We see from the 2024 report that for every billion miles travelled by a vehicle, there is a 4.7 fatality rate (Department for Transport, 2025). This is a sharp decline of 3% from the previous year(Department for Transport, 2025). There is also a lingering factor that for every casualty recorded, at least a car (as a form of transportation) is reported to be involved in the collision. The most common types of casualties in Great Britain are spread across car occupants, pedestrians, motorcyclists and pedal cyclists, with 43%, 26%, 21% and 5% respectively(Department for Transport, 2025). Although motorcyclist and pedestrian fatalities showed an increase in 2024 from the previous year, there was a decrease in fatalities in pedal cyclists and car occupants within the same period (Department for Transport, 2025).

Although reports show a steady decline in fatalities year in year out, this report takes a nosedive into the reported Road Safety Data - Collisions - 2024 through a data-driven approach, trying to understand the underlying factors in collisions.

Data-driven approaches to identifying hidden factors in collision patterns have been shown to enable targeted interventions with significantly higher effectiveness than broad safety programs (Rolison et al., 2018; Depaire et al., 2008). Through this data, we try to analyse the casualties, which group they are most common in, the type of vehicles involved, and other factors that might serve as major contributions to collisions. Finding hidden factors can serve as a benchmark to know what measures can be put in place to ensure that casualties are drastically reduced on public roads, as this deals with collisions and fatalities on public roads.

Chapter 2

Dataset - Road Safety Data - Collisions - 2024

2.1 Aim of this report

This analysis aims to discover what the data says about collisions and the factors involved in them for the year 2024 in Great Britain. This report will utilise the dataset to conduct general data analysis and perform supervised and unsupervised training.

2.2 Source of data

The data used in this analysis are open-source data freely available on the GOV.UK website. This data is encoded as textual strings, and another dataset was provided to decode the values. The data is presented in tabular form and does not include values for previous years; only 2024 is included. The data is stored in a two-dimensional array of 183514 x 32.

2.3 Data Analysis

The first thing was to look at the dataset and look for columns we would not need in this report, and drop those columns. We then assigned the columns needed to a new variable. These are the list of the variables dropped : *collision_index, collision_ref_no, vehicle_reference, towing_and_articulation, vehicle_manoeuvre_historic, vehicle_location_restricted_lane_historic, skidding_and_overturning, journey_purpose_of_driver_historic, engine_capacity_cc, propulsion_code, driver_imd_decile, lsoa_of_driver, driver_distance_banding, vehicle_left_hand_drive, generic_make_model, escooter_flag, collision_year.*

These variables were dropped as they will not be useful to the analysis we wish to carry out on the dataset. The next step was to clean the data for *'hit_object_in_carriageway'* and *'hit_object_off_carriageway'*; missing data were replaced with 'none.*'* Also, for *'age_of_driver,'* missing data were replaced with the mean of the ages available.

The analysis used a proportion on the scale of 0 to 1, with 0 meaning it never happened within a specific group, while 1 means it makes up the entire group.

2.3.1 Manoeuvres and Drivers' Gender

An analysis is done on the relationship between where a collision occurs, the kind of manoeuvres performed by the driver in that location, and the gender of the driver involved in that maneuver. To display this, the normalised stack bar chart (Figure 2.1) was used to show the proportional manoeuvres by junction location. This chart is necessary so that site-specific risks can be identified. There are 10 manoeuvres used in this analysis.

From this, we can see that the proportion of complex manoeuvres like turning right is higher at 'entering main road' than at any other junction location, but that same manoeuvre is almost at zero at *'not at or within 20 meters of junction.'* For this, we made use of the column 'junction_location' and *'vehicle_manoeuvre.'*

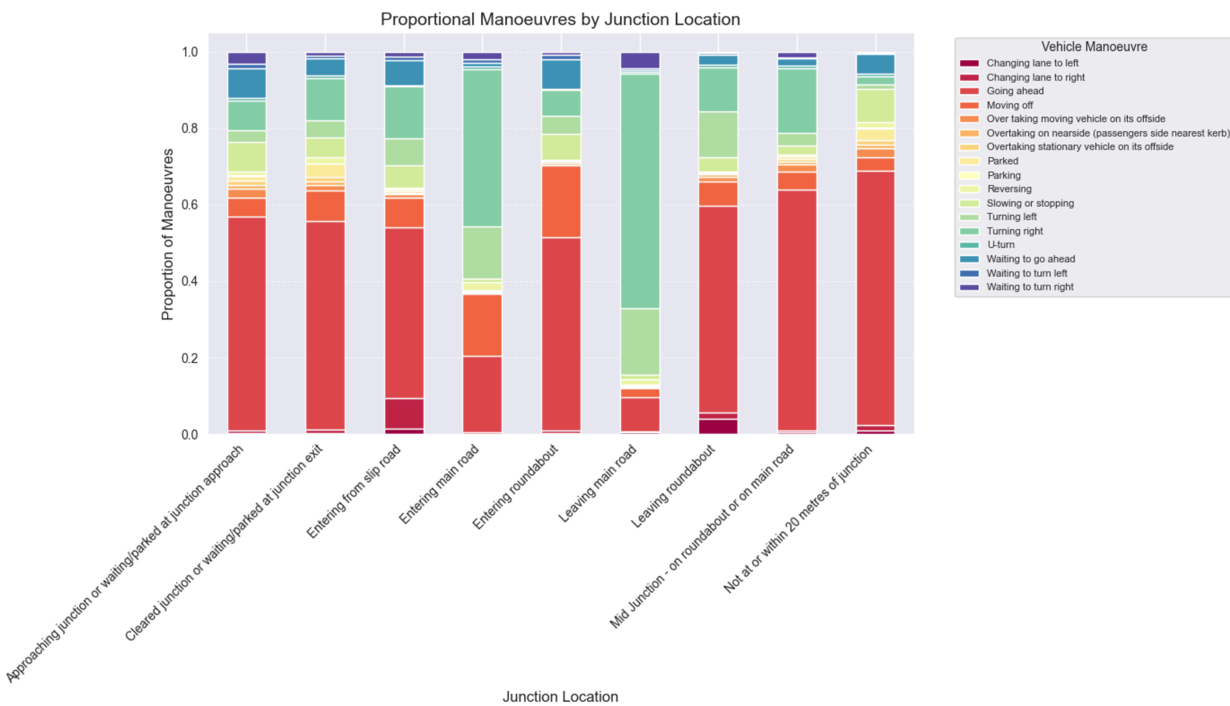

Figure 2.1 Proportional manoeuvres by junction location

The next aspect was to look at the count of all manoeuvres. Figure 2.2 provides a clear comparison of collisions by gender. From the Visualisation, the following can be derived: men are more involved in more collisions than women, and the table shows the magnitude of those collisions for every manoeuvre.

While ‘going ahead’ is the most frequent manoeuvre for both genders, the greatest male-to-female ratio (6.5:1) is *‘overtaking moving vehicle on its offside’*.

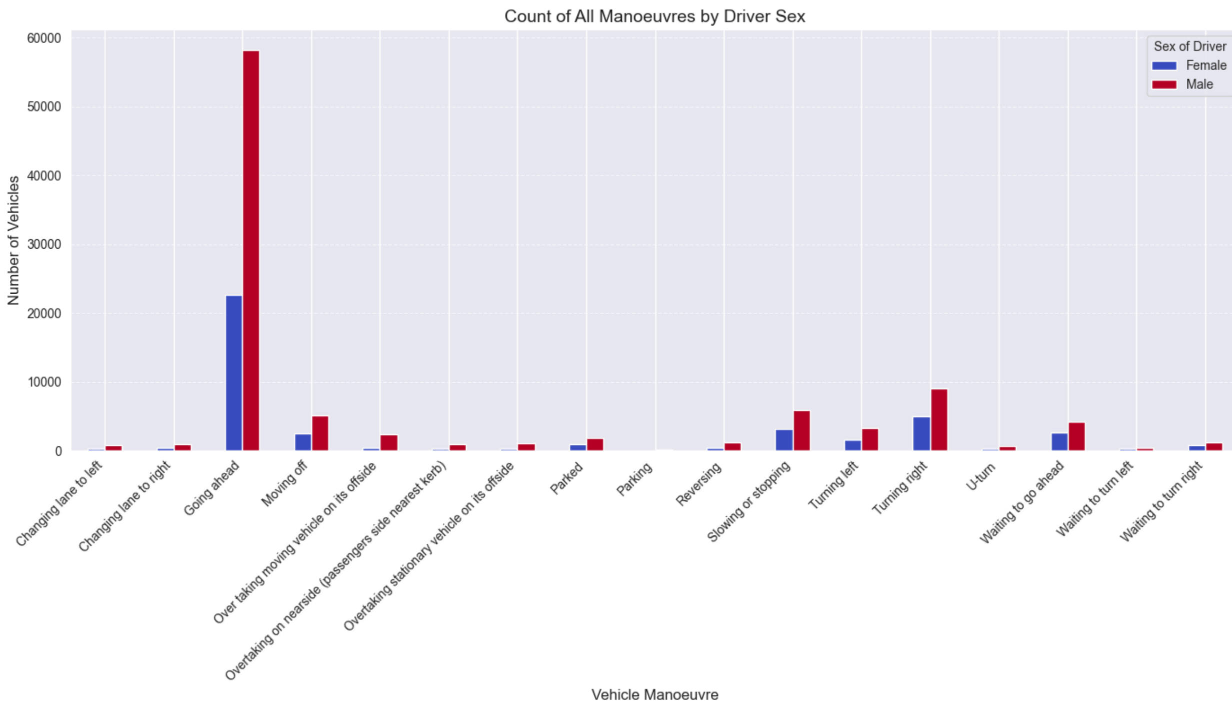

Figure 2.2 Manoeuvres by Driver Sex

2.3.2 Drivers' age and Vehicle profile

This analysis looks and the driver's age and the type of vehicle involved in the collision. It looked at the mean/median and its spread across various vehicle types. In Figure 2.3, the youngest drivers are often riding a motorcycle 125cc and under. They also have a median of 27 and a mean age of 29.9. The oldest drivers are mostly driving a bus or a coach, with a median age of 49 years.

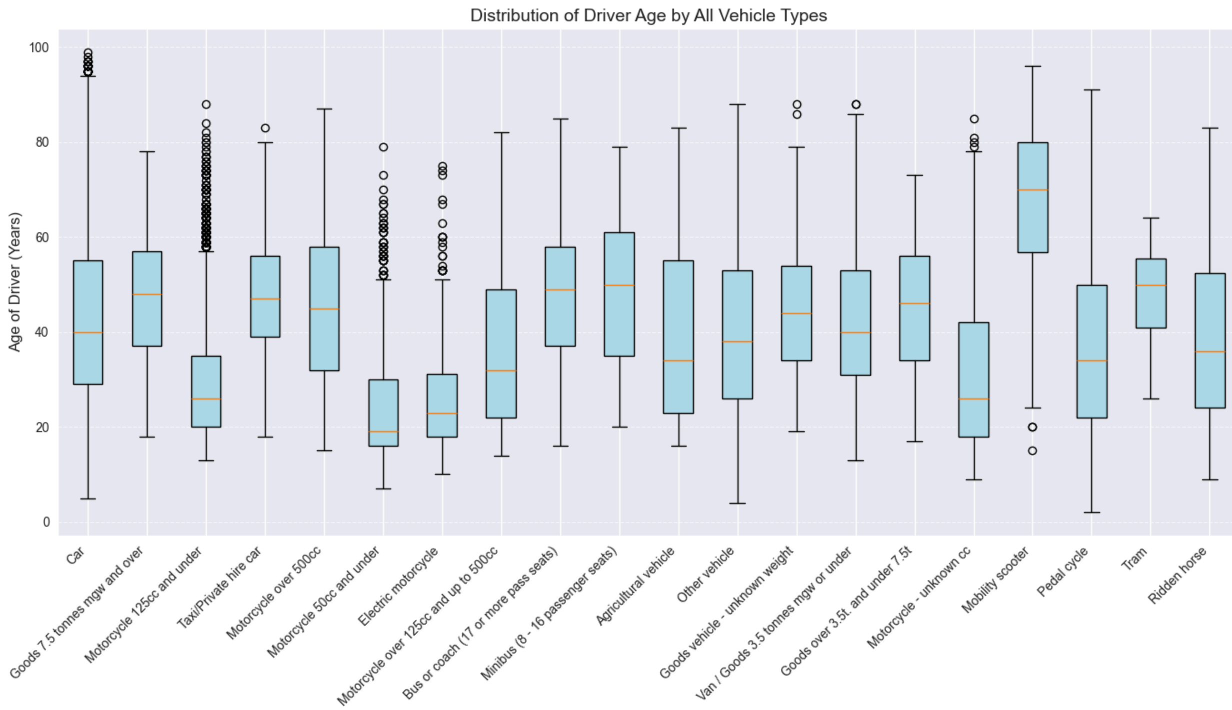

Figure 2.3 Distribution of Driver Age by Vehicle Types

Figure 2.4 shows the proportional breakdown of the first point of contact in a collision across all age bands as stipulated in the dataset. By showing the internal proportions of impact types within each age band, the chart highlights age-specific vulnerability. For example, the 16 -20 age band might show a higher proportion of Rear impacts (suggesting tailgating/inattention), while an older age band might show a higher proportion of Side impacts (suggesting difficulties judging speed/distance when turning).

Impact of front is the most common across all age bands, but it's proportionally highest for the age band 16-20. This accounts for more than half (60.8%) of their collisions. Also, for the age band 16-20, which includes younger drivers, the analysis shows that the ratio of front to back impact is 6.8:1, showing that they are more likely to have a front impact than an impact from the back by over 6 times.

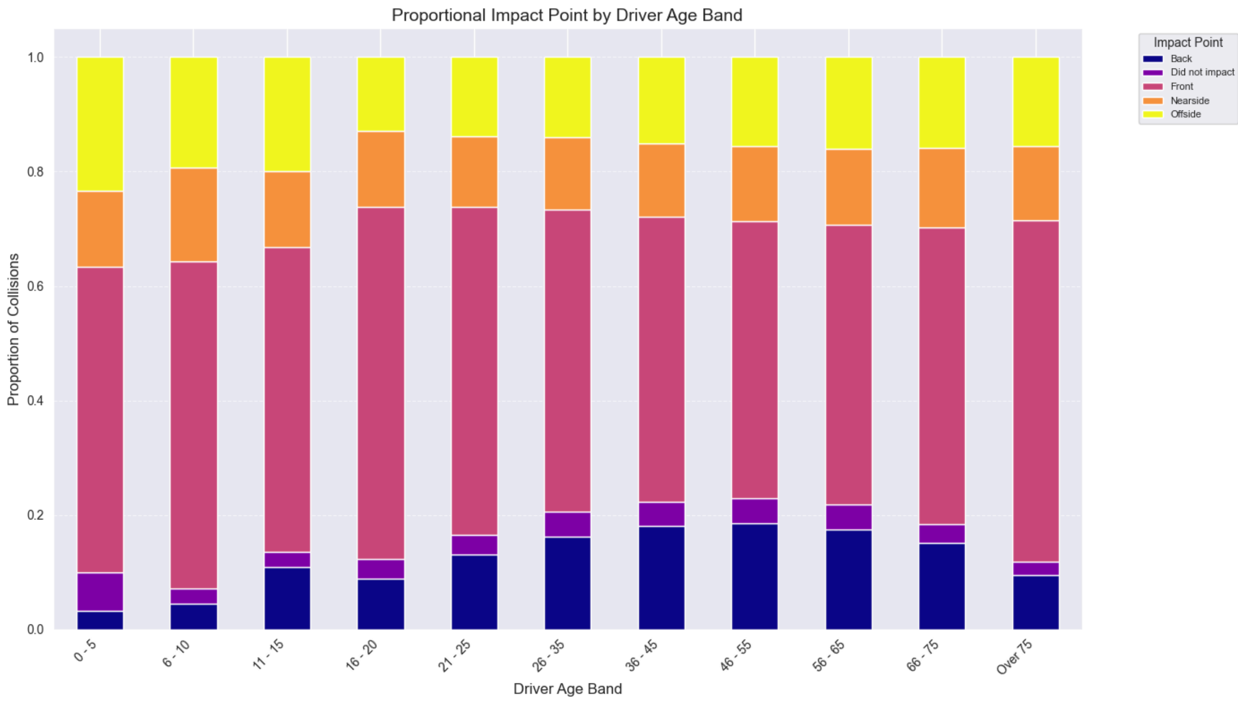

Figure 2.4 Proportional impact point by Driver Age Band

Chapter 3

3.1 Unsupervised Learning

3.2 K-MEANS Clustering

3.3 Overview

This analysis looks to find natural, distinct groups of drivers/vehicles that share similar characteristics in a collision. This analysis seeks to create clusters with different features and see what the accident patterns are amongst members of that cluster, so that a tailored approach can be generated to reduce collisions in that cluster.

K-means is one of the widely used clustering methods across various industries, especially transportation safety. Although there are various algorithms to use, K-means remains the benchmark due to simplicity, speed and interpretability (Jain, 2010).

For this Dataset, variables like 'vehicle_type', 'junction_location' and other categorical data were used with a One-Hot encoding step, which converted all categorical variables into numerical associations from which we could perform the k-means clustering on. To prevent columns with a large variety of data ('age_of_driver' with data points from 0 -99) from dominating the proximity calculation over columns with binary information (0's and 1's), a standard scaling was performed on features in that category, so all features are treated equally.

3.4 Results

The standardised feature values range from -2.588 to +2.554, which shows a large discrimination between clusters, with locations (from and to) and direction (North, east, west, south) providing the strongest separating power. Table 3.2 shows the 4 clusters and how their features work out (Because of the size of the table, only the first 23 variables are displayed. The Complete table can be found in Appendix 1).

	0	1	2	3	STD (Standard deviation)
driver	0.065	-0.065	0.221	-0.016	0.113377791
vehicle	0.01	0.014	-0.07	0	0.04105808397
Bus or coach (17 or more pass seats)	0.018	-0.015	0.248	-0.1	0.1380255319
Car	-0.01	0.018	0.038	-0.048	0.03644618089
Electric motorcycle	-0.042	0.015	-0.042	0.015	0.03525632444
Goods 7.5 tonnes mgw and over	0.096	-0.015	-0.05	-0.004	0.05973801649
Goods over 3.5t. and under 7.5t	-0.024	-0.024	-0.024	0.074	0.0476571086
Goods vehicle - unknown weight	-0.034	-0.034	-0.034	0.105	0.06758698099
Minibus (8 - 16 passenger seats)	0.136	-0.024	-0.024	-0.024	0.07496132336
Motorcycle - unknown cc	-0.042	0.015	0.076	-0.042	0.05454194797
Motorcycle 125cc and under	-0.008	-0.006	0.036	0	0.01909046459
Motorcycle 50cc and under	0.09	-0.032	0.037	-0.01	0.0491199335
Motorcycle over 125cc and up to 500cc	-0.01	-0.007	-0.07	0.055	0.05149618226
Motorcycle over 500cc	0.035	-0.002	-0.128	0.043	0.07996368275
Other vehicle	0.072	-0.019	-0.059	0.021	0.05384116508

Table 3.2 K-Mean Cluster Profiles

3.4.1 Cluster 0:

This cluster represents northern heavy vehicle transit accidents. Also, there is a strong north travel (1.822) and south departure points (1.885), involving heavy goods vehicles over 7.5 tonnes (0.096) and minibuses (0.136) operated by experienced drivers aged 46-65 on personal or leisure journeys. These accidents involve active manoeuvres such as overtaking stationary vehicles (0.105) and turning left (0.079), with outcomes including run-off-road incidents into ditches (0.151) and strikes against road signs (0.157).

From the analysis, this cluster suggests a specific demographic of experienced drivers operating large vehicles on main carriageways, presenting moderate-high severity on roads leading north.

3.4.2 Cluster 1:

The analysis in this cluster captures routine commuter traffic moving along the east-west route, with dominant directional features including north-east (0.250), north-west (0.252), and west (0.309) travel patterns. This cluster has no values exceeding 1.0; this is the largest cluster representing baseline accident scenarios involving standard cars (0.018), pedal cycles (0.048), and younger drivers aged 16-25 (0.032-0.035) engaged in forward travel (0.033) on main carriageways. This cluster represents typical commuter accidents at moderate severity but high volume.

3.4.3 Cluster 2:

The analysis in this cluster reveals a critical pedestrian safety emergency, displaying the most extreme feature concentrations in the entire dataset. Vehicles positioned on footways and pavements show an exceptionally high scaled value (2.554), while the corresponding value for main carriageway positioning is (-2.588), definitively confirming these vehicles are not in appropriate locations. The parked vehicle status appears consistently across three different feature measurements (0.598, 0.624, 0.536), with accidents involving buses and coaches (0.248), reversing manoeuvres (0.228), and impacts with walls or fences (0.271). This cluster shows a strong association with elderly drivers aged 66-75 (0.166) and over 75 (0.090), alongside huge missing data for journey purpose (0.238), suggesting investigation challenges.

3.4.4 Cluster 3:

The analysis in this cluster represents the flow of vehicles going to the south. It shows strong south travel (1.189) from northern departure points (1.154), involving vans and light goods vehicles under 3.5 tonnes (0.124) and medium-weight goods vehicles (0.105) operated predominantly by male drivers (0.081) on work journeys (0.073). These accidents occur during forward travel (0.135) on main carriageways with specific infrastructure strike patterns, including bridge sides (0.123) and central roundabout islands (0.109).

From this, we can infer that there are height and width awareness issues among commercial drivers using the southern routes and roundabouts.

Chapter 4

4.1 Supervised Learning

4.2 Decision Tree Classifier

The supervised approach used in this analysis is the decision tree approach. Here, the approach aims to predict the point of initial contact in a collision. In Figure 4.1, we see the model's classification results for predicting the first point of contact in a collision.

```
--- Detailed Classification Report ---
                precision    recall  f1-score   support

          Back       0.43      0.20      0.27        15
         Front       0.00      0.00      0.00         3
      Nearside       0.74      0.91      0.81       251
       Offside       0.09      0.02      0.04        42
Did not impact       0.06      0.03      0.04        35

      accuracy                           0.67       346
     macro avg       0.26      0.23      0.23       346
  weighted avg       0.57      0.67      0.61       346
```

Figure 4.1 classification report

The model was tested on 346 accidents and correctly predicted the impact location 67% of the time. The model looked at many different factors, including what type of vehicle was involved, who was driving, what direction they were going, and the manoeuvre happening when the accident happened. The two most important factors for making predictions were the type of vehicle involved and the characteristics of the driver (age, gender), as seen in Figure 4.2, which together were much more useful than any other information the model had.

When examining which factors mattered most, vehicle type was by far the most important with a score of 0.14, followed by driver characteristics at 0.07, meaning these two pieces of information were the best clues for figuring out where a vehicle would first be hit in a collision.

The biggest problem with this model is that it works well for one type of accident but fails badly for the others. Out of the 346 test cases, 251 were nearside impacts, meaning the vehicle was hit on the passenger side nearest the curb, and the model predicted these correctly 91% of the time with good accuracy. However, for the other types of impacts, the model performed terribly.

```
--- Top 10 Feature Importances for Impact Prediction ---
vehicle                                 0.142922
driver                                  0.078461
West                                    0.046030
Reversing                               0.045629
Nearside and rebounded                  0.035647
Not known or not requested              0.034922
Wall or fence                           0.031843
Male                                    0.030713
Offside on to centrl res + rebounded    0.023316
Taxi/Private hire car                   0.023000
```

Figure 4.2 Top 10 important features

To make this model actually useful in the real world, several major improvements would be needed. First, the training data needs to be balanced so the model sees enough examples of each type of impact, especially front impacts, which had only three examples.

Chapter 5

Reflections on methods used for analysis

I’ll start with the supervised learning part of this assessment since it’s the last thing I worked on. The model performs poorly overall. When you ignore the one category it got right (nearside impacts), the accuracy drops to just 23%. A good model should work well for all categories, not just the most common one. Decision trees struggle badly when the data is unbalanced, meaning when one category appears much more often than others, which is exactly what's happening here with 251 nearside impacts out of 346 total cases. I think a better model to use would be a random forest. In retrospect, it seems that would align more with what I’m trying to achieve with this analysis.

I’m happy with how the assessment for K-means turned out for the unsupervised learning part of this assessment. The data has the right structure for K-means, especially when categorical data were encoded. The only issue I can raise with this is that the clusters do not seem to be equal. Also, I should be validating why I used K = 4 and validating it against 3 and 5 to ensure that I picked the optimal number, as that was selected at random, and since the results were fairly optimal, I left it at that.

Overall, the methods used show that sometimes you have to try various methods to see the one that fits more with what you are trying to achieve. My biggest takeaway is that, for a successful model training, one must be intentional about data collection and validation from the beginning.

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Environment

Language: Python 3.12.0

Packages used:

- Pandas
- Numpy
- Matplotlib
- sklearn

IDE: Phycharm

Dataset derived from: https://www.gov.uk/government/statistics/road-safety-data

Appendix 1: K-Means Cluster result

	0	1	2	3	STD
driver	0.065	-0.065	0.221	-0.016	0.113377791
vehicle	0.01	0.014	-0.07	0	0.04105808397
Bus or coach (17 or more pass seats)	0.018	-0.015	0.248	-0.1	0.1380255319
Car	-0.01	0.018	0.038	-0.048	0.03644618089
Electric motorcycle	-0.042	0.015	-0.042	0.015	0.03525632444
Goods 7.5 tonnes mgw and over	0.096	-0.015	-0.05	-0.004	0.05973801649
Goods over 3.5t. and under 7.5t	-0.024	-0.024	-0.024	0.074	0.0476571086
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Motorcycle 50cc and under	0.09	-0.032	0.037	-0.01	0.0491199335
Motorcycle over 125cc and up to 500cc	-0.01	-0.007	-0.07	0.055	0.05149618226
Motorcycle over 500cc	0.035	-0.002	-0.128	0.043	0.07996368275
Other vehicle	0.072	-0.019	-0.059	0.021	0.05384116508

Pedal cycle	0.073	0.048	-0.094	-0.094	0.09118951039
Taxi/Private hire car	-0.082	0.029	0.055	-0.034	0.06204303652
Van / Goods 3.5 tonnes mgw or under	-0.096	-0.029	-0.017	0.124	0.09105004105
Changing lane to right	-0.103	0.026	-0.054	0.037	0.07048935969
Going ahead	-0.042	0.033	-0.365	0.135	0.2240936402
Moving off	-0.012	-0.026	0.193	-0.034	0.1006366329
Over taking moving vehicle on its offside	0.049	0.025	-0.159	-0.003	0.0963102647
Overtaking on nearside (passengers side nearest kerb)	-0.08	0.01	0.105	-0.021	0.07461885448
Overtaking stationary vehicle on its offside	0.105	-0.035	0.057	-0.023	0.06050930542
Parked	-0.08	-0.08	0.598	-0.08	0.3140713295
Parking	-0.034	0.001	0.11	-0.034	0.06440202801
Reversing	-0.032	-0.02	0.228	-0.05	0.1224251044
Slowing or stopping	0.059	-0.003	0.087	-0.072	0.06561854117
Turning left	0.079	0.015	0.129	-0.14	0.10997495
Turning right	0.046	-0.049	0.14	0.001	0.07252077017
U-turn	-0.048	0.001	0.054	0.001	0.04020000996
Waiting to go ahead	-0.003	-0.026	0.013	0.048	0.02884857573

Waiting to turn left	-0.024	0.025	-0.024	-0.024	0.02670205985
Waiting to turn right	-0.034	0.036	-0.034	-0.034	0.03806835957
North	1.822	-0.384	0.153	-0.425	0.9741827056
North East	-0.211	0.25	-0.113	-0.312	0.2613006753
North West	-0.229	0.252	-0.12	-0.302	0.2633716169
Parked	-0.084	-0.084	0.624	-0.084	0.3280414608
South	-0.44	-0.433	-0.127	1.189	0.7459454223
South East	-0.279	0.209	-0.021	-0.232	0.2364867958
South West	-0.281	-0.266	-0.03	0.713	0.4484352487
West	-0.375	0.309	-0.002	-0.382	0.3499058061
North	-0.404	-0.433	-0.104	1.154	0.7191791642
North East	-0.3	-0.24	-0.015	0.665	0.4256484836
North West	-0.286	0.201	0.048	-0.247	0.2450206863
Parked	-0.032	-0.08	0.536	-0.08	0.277698194
South	1.885	-0.399	0.098	-0.409	1.005912985
South East	-0.134	0.2	-0.047	-0.291	0.2156796498
South West	-0.314	0.285	-0.095	-0.327	0.3049400527
West	-0.364	0.292	0.05	-0.38	0.3433925073
Cycle lane (on main carriageway)	-0.048	0.051	-0.048	-0.048	0.05381635439
Cycleway or shared use footway (not part of main carriageway)	-0.054	0.035	0.037	-0.054	0.05262050848
Footway (pavement)	-0.342	-0.342	2.554	-0.342	1.341565951

Lay-by or hard shoulder	-0.059	-0.059	0.441	-0.059	0.2316164934
On main carriageway (not in restricted lane)	0.355	0.338	-2.588	0.36	1.518739169
Tram or Light rail track	-0.048	-0.048	0.36	-0.048	0.188976189
Cleared junction or waiting/parked at junction exit	0.028	-0.027	0.278	-0.098	0.151162082
Entering from slip road	-0.076	0.002	0.182	-0.045	0.1094045887
Entering main road	0.068	0.002	-0.007	-0.041	0.04336806065
Entering roundabout	0.058	-0.027	-0.159	0.094	0.1113778527
Leaving main road	0.019	0.012	0.189	-0.126	0.1212339243
Leaving roundabout	0.111	-0.052	-0.004	0.037	0.06322332588
Mid Junction - on roundabout or on main road	0.074	0.023	-0.199	0.005	0.1224238966
Not at or within 20 metres of junction	-0.16	0.045	-0.142	0.078	0.1309080814
Bollard or refuge	0.129	0.003	-0.128	-0.022	0.1038386479
Bridge (side)	-0.049	-0.021	-0.107	0.123	0.0982345201
Central island of roundabout	0.031	-0.016	-0.2	0.109	0.1320096666
Kerb	-0.072	0.043	0.13	-0.104	0.1050501607

Open door of vehicle	0.051	-0.013	0.076	-0.042	0.0502450643
Other object	-0.009	-0.05	0.073	0.071	0.05569028053
Parked vehicle	-0.027	-0.012	0.09	-0.003	0.04936210266
Previous accident	-0.003	0.011	0.013	-0.027	0.01828090157
Road works	0.092	-0.018	-0.087	0.022	0.07252718056
Nearside	-0.084	-0.038	0.126	0.067	0.09013018358
Nearside and rebounded	-0.071	0.013	0.062	-0.012	0.05433867837
Offside	0.033	0.035	0.056	-0.117	0.07746950104
Offside - crossed central reservation	-0.01	-0.029	-0.026	0.076	0.04773421806
Offside and rebounded	0.003	0.041	-0.134	-0.019	0.07978027284
Offside on to central reservation	0.046	0.028	-0.161	-0.006	0.0971530956
Offsicentralto central res + rebounded	0.088	-0.015	-0.128	0.037	0.09105004105
Straight ahead at junction	0.082	-0.023	-0.042	0.016	0.05179859096
Central crash barrier	0.043	-0.01	-0.135	0.059	0.08800888335
Entered ditch	0.151	-0.007	-0.092	-0.034	0.1003670112
Lamp post	-0.033	0.023	0.031	-0.041	0.03737391176
Near/Offside crash barrier	-0.014	0.012	-0.128	0.046	0.07829010725
Other permanent object	-0.019	-0.019	0	0.05	0.03117040247

Road sign or traffic signal	0.157	-0.033	-0.143	0.039	0.1217740514
Submerged in water	-0.034	0.001	-0.034	0.035	0.03402854899
Telegraph or electricity pole	-0.055	0.023	-0.053	0.014	0.04515905355
Tree	-0.016	0.047	-0.059	-0.056	0.05310780894
Wall or fence	-0.123	-0.005	0.271	-0.046	0.1625828517
Did not impact	-0.01	0.022	-0.094	0.008	0.05498246218
Front	0.098	-0.047	0.057	0.005	0.05669876217
Nearside	0.008	0.001	0.005	-0.009	0.006984128029
Offside	-0.065	0.033	-0.123	0.034	0.08245776271
Education and educational escort	0.056	0.017	-0.025	-0.055	0.04765621082
Emergency vehicle (blue light) on response	-0.048	0.001	-0.048	0.05	0.04824205645
Journey as part of work	-0.019	-0.023	-0.034	0.073	0.04847358943
Not known or not requested	-0.102	0.047	0.238	-0.145	0.1668386734
Personal business or leisure	0.148	-0.079	-0.123	0.126	0.1318536112
Male	-0.108	0.018	-0.104	0.081	0.09748390503
16 - 20	-0.024	0.032	-0.102	0.001	0.06129992413
21 - 25	-0.066	0.035	-0.158	0.047	0.1019442731
26 - 35	0.063	0.024	0.014	-0.092	0.06416010117
36 - 45	-0.127	0.02	-0.042	0.058	0.08391232873

46 - 55	0.084	-0.068	0.024	0.072	0.06255423321
56 - 65	0.085	-0.051	0.101	0.001	0.06430088437
66 - 75	-0.006	-0.002	0.166	-0.073	0.09528315403
Over 75	0.013	-0.018	0.09	-0.016	0.0462185158